Hashing

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What Is a Hash Function?

A **hash function** maps keys to slots in a hash table of size m:

$$h : \mathcal{U} \to \{0, 1, \ldots, m-1\}.$$

What makes a good hash function?

- **Uniform**: each key k is equally likely to map to any slot.

$$\Pr[h(k) = j] = \frac{1}{m} \quad \text{for all } j.$$

- **Fast**: $O(1)$ to compute.
- **Deterministic**: same key always maps to same slot.

Key Types

- **Integers**: typically 32-bit or 64-bit.

$$k \in \{0, 1, \ldots, 2^{32} - 1\}$$

- **Byte strings**: a short vector of non-negative integers of bounded size (e.g. strings, IP addresses, DNA sequences).

$$k = (k_0, k_1, \ldots, k_{r-1}), \quad k_i \in \{0, \ldots, 255\}$$

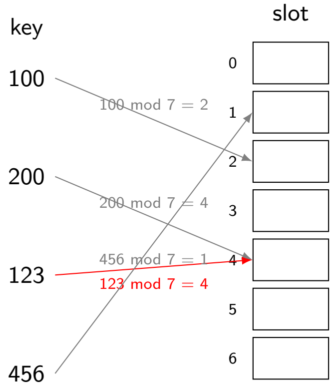
Division Method

$$h(k) = k \bmod m$$

- **Simple**: one modulo operation.
- **Works well**: when m is a prime not too close to a power of 2.
- **No theoretical guarantee** in general.

Division Method: Example

$m = 7$ (prime), keys $= 100,\ 200,\ 123,\ 456.$



200 and 123 collide at slot 4.

Multiplication Method

Choose a constant $0 < A < 1$. Then:

$$h(k) = \lfloor m \cdot (kA \bmod 1) \rfloor$$

where $kA \bmod 1$ denotes the **fractional part** of kA.

Steps:

1. Multiply k by A: get kA.
2. Keep only the fractional part: $\{kA\} = kA - \lfloor kA \rfloor \in [0, 1)$.
3. Scale by m and floor.

- Works for any m (not just primes).
- Knuth recommends $A = \dfrac{\sqrt{5} - 1}{2} \approx 0.6180$.

Multiplication Method: Example

$m = 8$, $A = 0.618$, keys $= 100,\ 200,\ 123,\ 456.$

k	kA	$\{kA\}$	$h(k) = \lfloor 8 \cdot \{kA\} \rfloor$
100	61.800	0.800	$\lfloor 6.400 \rfloor = 6$
200	123.600	0.600	$\lfloor 4.800 \rfloor = 4$
123	76.014	0.014	$\lfloor 0.112 \rfloor = 0$
456	281.808	0.808	$\lfloor 6.464 \rfloor = 6$

100 and 456 collide at slot 6. The other keys are spread across different slots.

Multiply-Shift Method

Setting: w-bit integer keys, $m = 2^l$ slots for some $l < w$.

Choose a **random odd** w-bit integer a. Then:

$$h(k) = \left\lfloor \frac{(a \cdot k) \bmod 2^w}{2^{w-l}} \right\rfloor$$

Equivalently: multiply k by a, keep the lower w bits, then take the **top l bits**.

- **Fast**: one multiplication + one bit shift.
- Replaces the expensive modulo with bit operations.
- Theoretically motivated: the top l bits mix the input well.

Multiply-Shift: Illustration

$w = 8$ bits, $l = 3$, $m = 2^3 = 8$ slots, $a = 203 = 11001011_2$ (odd).

Key $k = 5 = 00000101_2$:

$$a \times k = 203 \times 5 = 1015$$

$$1015 \bmod 256 = 247 = 11110111_2$$

$$\overbrace{1\ \ 1\ \ 1\ \ 0}^{\text{top } l = 3 \text{ bits} = 111_2 = 7}\ \ \underbrace{1\ \ 1\ \ 1\ \ 1}_{\text{lower } w - l = 5 \text{ bits (discarded)}}$$

$h(5) = 111_2 = 7.$

Multiply-Shift: More Examples

$w = 8$, $l = 3$, $m = 8$, $a = 203$.

k	$203k \bmod 256$	Binary	$h(k)$ (top 3 bits)
3	97	01100001	$011 = 3$
5	247	11110111	$111 = 7$
7	141	10001101	$100 = 4$
10	238	11101110	$111 = 7$

Keys 5 and 10 collide at slot 7.

A different choice of a gives a different hash function — this is the basis of **universal hashing**.

The Problem with Fixed Hash Functions

Any fixed h has a worst case.

- An adversary who knows h can craft n keys that all map to the same slot.
- Result: every lookup takes $O(n)$ time.

Fix: choose h *at random* from a family $\mathcal{H}$.

The adversary must choose keys *before* seeing which h was drawn — collisions become unlikely.

2-Universal Family

Definition: A family $\mathcal{H}$ of functions $h : \mathcal{U} \to \{0, \ldots, m-1\}$ is **2-universal** if for any two distinct keys $x, y \in \mathcal{U}$:

$$\Pr_{h \sim \mathcal{H}}\big[h(x) = h(y)\big] \leq \frac{1}{m}.$$

Intuition: a random function would collide with probability exactly $1/m$. 2-universality matches this ideal.

Consequence: Expected Collisions

Fix any n keys $x_1, \ldots, x_n$. Draw $h \sim \mathcal{H}$.

Let $C_{ij} = \mathbf{1}[h(x_i) = h(x_j)]$. By 2-universality:

$$\mathbb{E}[C_{ij}] \leq \frac{1}{m} \quad \text{for } i \neq j.$$

Total expected collisions:

$$\mathbb{E}\left[\sum_{i<j} C_{ij}\right] \leq \binom{n}{2} \cdot \frac{1}{m}.$$

With $m = n$: at most $\frac{n-1}{2} < \frac{n}{2}$ expected collisions — $O(1)$ expected chain length.

Simple construction

Let p be a prime with $p \geq |\mathcal{U}|$. Choose $a \in \{1, \ldots, p-1\}$ and $b \in \{0, \ldots, p-1\}$ uniformly at random. Define:

$$\boxed{h_{a,b}(k) = \big((ak + b) \bmod p\big) \bmod m}$$

Theorem (Carter–Wegman 1979): $\mathcal{H} = \{h_{a,b}\}$ is 2-universal.

What Is a Field?

A **field** is a set with $+$ and $\times$ where every nonzero element has a multiplicative inverse.

Examples: $\mathbb{R}$, $\mathbb{Q}$. And, for prime p — $\mathbb{Z}_p = \{0, 1, \ldots, p-1\}$ with arithmetic mod p.

$$
\begin{array}{c|l}
\mathbb{Z}_2 & \{0,1\}: \quad 1^{-1} = 1 \\
\mathbb{Z}_3 & \{0,1,2\}: \quad 1^{-1} = 1, \quad 2^{-1} = 2 \\
\mathbb{Z}_5 & \{0,1,2,3,4\}: \quad 1^{-1} = 1, \quad 2^{-1} = 3, \quad 3^{-1} = 2, \quad 4^{-1} = 4
\end{array}
$$

Key property we use: if $a \neq 0$ in $\mathbb{Z}_p$, then $ax = b$ has the unique solution $x = a^{-1}b$.

$\mathbb{Z}_5$: Multiplication Table

$\times$	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

Every nonzero row is a permutation of $\{1, 2, 3, 4\}$ — every nonzero element appears exactly once, so each has a unique inverse:

$$1^{-1} = 1, \quad 2^{-1} = 3, \quad 3^{-1} = 2, \quad 4^{-1} = 4.$$

Proof: Bijection

Fix distinct x, y. Let $r = (ax + b) \bmod p$, $s = (ay + b) \bmod p$.

Claim: for any target (r, s) with $r \neq s$, there is a *unique* (a, b) producing it.

Proof of claim: subtracting the two equations gives $a(x - y) \equiv r - s \pmod{p}$.

Since $\mathbb{Z}_p$ is a field and $x - y \neq 0$, the inverse $(x - y)^{-1}$ exists and is unique, so:

$$a = (r - s)(x - y)^{-1} \bmod p \quad (\text{unique, and } \neq 0 \text{ since } r \neq s).$$

Then $b = r - ax \bmod p$ is uniquely determined.

So $(a, b) \mapsto (r, s)$ is a bijection: (r, s) is **uniform** over all $p(p - 1)$ pairs with $r \neq s$.

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Conclusion

$h(x) = h(y)$ iff $r \equiv s \pmod{m}$.

For each r, at most $\dfrac{p-1}{m}$ values of $s \neq r$ satisfy $s \equiv r \pmod{m}$.

So the number of "bad" pairs (r, s) is at most $p \cdot \dfrac{p-1}{m}$, giving:

$$\Pr[h(x) = h(y)] \leq \frac{p(p-1)/m}{p(p-1)} = \frac{1}{m}. \qquad \square$$